

sensors. It must be assumed by the designer here that cell temperature (either internal or ambient) will affect adjoining cells as well, and so overall temperature of the battery can be estimated by using only a few temperature sensors.

Ideally, once the battery is fabricated, a temperature distribution analysis must be carried out while the battery goes through at least a single charge and discharge cycle. Such an analysis will show the hot spots of the battery and would help the BMS installer to decide on points of installation of the temperature sensors.

Simpler BMSes either provide a single temperature sensor for the entire battery or do not provide any. A good and responsible BMS should have at least 4-5 temperature sensors. Typically, the temperature sensors used for BMSes are resistive temperature devices and sometimes thermocouple modules. Advanced BMSes may also provide ways to control the battery temperature by either using forced air/liquid cooling or heating up the battery using Peltier elements.

Controlling charging and discharging

Charging and discharging a battery of cells is the primary process that the battery goes through. Controlling the process of charging and discharging ensures that the battery remains healthy for the longest time possible. However, this kind of control is mostly an on-off type control.

Discharging depends mostly on the load connected to the battery and the BMS has no control over the load, and so it controls the discharge rate by limiting the discharge current. Similarly, the BMS also does not have any control on the charger capacity, and it only controls the charging rate by limiting the charging current. The BMS almost stays transparent to charge movement until the charge or discharge current reaches the specified maximum value, at which point it kicks-in and disconnects or reconnects

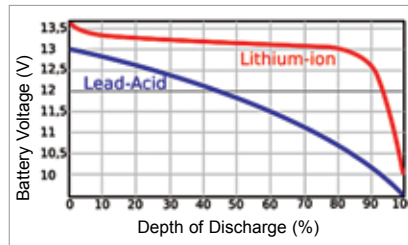


Fig. 2: Depth of discharge against OCV of Li-ion and lead-acid batteries (Source: <https://www.zwercat.nl/en>)

the load or source from the battery.

It is interesting to note that the maximum charging and discharging rates of a battery depend directly on its capacity. This capacity is typically expressed in the unit of ampere-hour (AH), which is mostly a glorified unit of charge (remember, $Q = I \times T$). Different batteries having the same nominal OCV can differ from each other based on their capacities. For example, a standard 12V battery is available in various capacities, such as 7AH, 35AH, 50AH, 80AH, 100AH, and so on. Battery cell capacities are typically specified in mAh, which is milli-ampere-hour.

Since the current capacity of the above batteries is different, this means the BMS used for each one of them must also be different. The BMS designer, therefore, must specify a current capacity against which the BMS will work. BMSes are always specified in terms of the typical voltage and maximum current that they can handle.

The BMS voltage is usually specified in terms of the series connection of the constituent cells. For example, a 3S Li-ion BMS means that it expects to be connected to a battery having 3 Li-ion cells connected in series. This means that when choosing a BMS for battery, the user must look at both the battery capacity and the application requirement, and choose a BMS such that:

Battery Current Capacity $\geq$ BMS Current Specification $\geq$ Application Current Requirements.

Some BMSes allow a single connection point for charging and

discharging. This is the simplest design, while some others allow different points for charging and discharging connections. The logic behind a single connection is typically lower price. However, with only one connection point, the battery can only be connected to either the load or the charger.

With two connection points, the BMS allows connecting the battery to the charger and to the load at the same time. This is the most common scenario in our day-to-day use. But the mobile phone BMS allows use of the battery to power the mobile phone while also being charged via its USB device port.

The BMS also needs to use different cut-off switches for the charge and discharge current paths of the battery. This is because the charging path is from the charger to the battery, while the discharging path is from the battery to the load, and a BMS typically uses a MOSFET (or other static switches) for connecting and disconnecting the battery from the external world.

These devices are unidirectional, so two different FETs have to be used. In this case, they are connected back-to-back and use each other's body diode to complete the charge and discharge paths. While this allows independent control of the charging and discharging current limit, the BMS still needs to provide different connection points, in case the battery has to be connected to the load and the charger simultaneously. If there is a single connection point, either of the current limits can lead to the port being disconnected, stalling the other process as well.

Even though we can provide two connection points, the battery itself can function only in one mode at a time, either charging or discharging. Let us understand this in some more detail. The typical voltage condition of a battery system would be such that:

Charging Voltage $>$ Battery Voltage $>$ Application Requirement

This must hold at all times, so that